MIL-C-29127A(SA)
2 October 1978
SUPERSEDING
MIL-C-29127(SA)
31 August 1973

MILITARY SPECIFICATION

CLOTH, TWILL, POLYESTER/COTTON (CREASE-RESISTANT FINISH)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for polyester and cotton blend twill fabrics treated with a crease-resistant finish.

1.2 Classification. The cloth shall be of one type in the following classes (see 6.2):

Class 1 - dyed
Class 2 - white

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-covered
MIL-T-43624 - Thread, Polyester, Spun

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580).

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials.

3.3.1 Fiber blend.

3.3.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate, semi-dull. The denier shall not be greater than 2.5 and the staple shall be approximately 1 1/2 inches (3.8cm) in length.

3.3.1.2 Cotton fiber. The cotton shall be combed.

3.3.2 Yarn. The warp and filling yarn shall be a blend of 50 (± 3) percent polyester and the remaining percentage cotton based on the dry weight on the de-sized and bleached cloth before treatment. The warp shall be a two-ply yarn and the filling shall be a singles yarn.

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3.4 Color.

3.4.1 Class 1, dyed. The color of the dyed cloth shall be as specified (see 6.2) and shall match the standard sample.

3.4.1.1 Sulfur content. The use of sulfur dyes or dyes containing elementary sulfur or sulfur compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.5.

3.4.1.2 Colorfastness. The finished cloth shall show colorfastness to laundering (after 3 cycles), crocking, perspiration and light equal to or better than the standard sample (see 6.3). When no standard sample is available, the cloth shall show a minimum of "good" colorfastness to laundering (after 3 cycles), perspiration, light and a Munsell value for crocking not lower than 8.5, when tested as specified in 4.5.

3.4.2 Class 2, white. The color of the class 2 finished cloth shall be white 3053 conforming to the standard sample when tested as specified. All cloth within any one contract shall be processed for whiteness in the same manner. The color shall be obtained with a chemical bleach, preferably with hydrogen peroxide or peroxygen chemicals. The chemical bleach may be supplemented with fluorescent optical brighteners which fluoresce to the blue region

3.4.2.1 Discoloration. The color shall show no more discoloration to light, perspiration and combined laundering and bleaching than that of the standard sample when tested as specified in 4.5.

3.4.3 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500°Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800°Kelvin.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

Table I - Physical requirements

Weight (min) oz/sq yd(g/m^2)	Yarns per inch (cm) (min)		Breaking strength lbs.(newton)(min)		Tearing strengt lbs.(newton)(min	
	Warp	Filling	Warp	Filling	Warp	Fillin
6.8 (231)	102(48)	50(20)	170(757)	85(378)	7.0(31)	5.0 (2

3.5.1 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

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3.5.2 Weave. The weave shall be a 2 up 1 down, right-hand twill.

3.6 Non-fibrous materials. The total non-fibrous material content of the finished cloth shall not exceed 6.0 percent when tested as specified in 4.5.

3.7 Finish. The cloth shall be desized, scoured, mercerized, heat set, bleached (Class 2 only) and treated with a resin to provide a crease resistant finish and hand equal in characteristics to that of the standard sample and shall meet the requirements specified herein. The resin finish shall be of a non-chlorine retentive type and the material shall show no loss of colorfastness or no more discoloration than the standard sample when tested as specified in 4.5.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.9 Shrinkage. The cloth shall be preshrunk so that it shall not shrink more than 2.0 percent in the warp or filling when tested as specified in 4.5. The preshrinking process shall not be identified by name or trademark either on the cloth or on the ticket or package.

3.10 Seam efficiency. The cloth shall have a seam efficiency of not less than 90 percent when tested as specified in 4.5.

3.11 Length and put-up. The minimum length of a continuous piece shall be 40 yards. The pieces shall be put up in rolls and ticketed as specified in PPP-P-1134.

3.12 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.13 Face identification. The face side of the cloth shall be identified by stamping on that side with the work "Face" at each end of the roll.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$) when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirement specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article of the finished cloth shall be visually inspected for shade, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.4.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.4.2.1 thru 4.4.2.3.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, (91.4m) the entire yardage in the roll shall be examined. When the total yardage in the roll exeeds 100 yards (91.4m), only 100 yards (91.4m) shall be examined. Defects as defined in Section I of FED-STD-4 which are clearly visible at normal inspection distance (3 feet) (0.9m) shall be scored and assigned demerit points as listed in 4.4.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.4), "C" for slubs and "B" for knots, shall be scored. No linear yard (increment of one yard (0.9m) on the measuring device of the inspection machine) from any one roll within the sample, shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) exceed 25 points. For determination of acceptability on the individual roll quality basis, the lot shall be unacceptable if two or more individual rolls exceed 38 points per 100 square yards ($83.6m^2$). If one roll exceeds the point level, a second sample of 20 rolls shall be selected for individual roll quality examination only. If one or more rolls exceed the value, the lot shall rejected. Point computation for the lot quality and individual roll quality shall be as follows:

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$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (7.6cm) or less in any dimension	- one point
For defects exceeding 3 inches (7.6cm), but not exceeding 6 inches (15.2cm) in any dimension	- two point
For defects exceeding 6 inches (15.2cm), but not exceeding 9 inches (22.9cm) in any dimension	- three points
For defects exceeding 9 inches (22.9cm) in any dimension	- four points

The following defects, when present, shall be scored four points for each yard (0.9m) in which they occur:

Baggy, ridgy or wavy cloth
Width of cloth less than specified
Characteristics of finish, hand and shade not equal to the standard sample
Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
Excessive neppiness

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade variation During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shaded variation side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, identification of preshrinkage process and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labelled or ticketed in accordance with the Textile Fiber Products Identification Act.

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4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking complies with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 continuous yards (2.7 continuous meters) full width of the finished cloth. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard (0.9m). The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test Method

Characteristic	Requirement paragraph	Test method
Polyester fiber; identification, denier and staple length	3.3.1.1	1/
Cotton fiber identification	3.3.1.2	1/
Combing of cotton fiber	3.3.1.2	1/
Polyester fiber content	3.3.2	2100 and 4.5.1
Cotton fiber content	3.3.2	2100 and 4.5.1
Yarn ply	3.3.2	Visual
Colorfastness to:		
Laundering (3 cycles)	3.4.1.2	5610 5/ 6/
Light	3.4.1.2	5660
Perspiration	3.4.1.2	5680
Crocking	3.4.1.2	5651
Discoloration to:		
Light (20 Standard Hours)	3.4.2.1	5660
Perspiration	3.4.2.1	5680
Combined laundering and bleaching	3.4.2.1	5605
Weight	3.5	5041
Yarns per inch (warp and filling)	3.5	5050
Breaking strength (warp and filling)	3.5	5100
Tearing strength (warp and filling)	3.5	5132
Weave	3.5.2	Visual 2/
Non-fibrous material	3.6	2610

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Table II - Test Method (cont'd)

Characteristic	Requirement paragraph	Test method
Finish		
Desized	3.7	1/
Scouring	3.7	1/
Bleaching agents and brighteners, Class 2 only)	3.7 and 3.4.2	1/
Heat set	3.7	1/
Crease resistant finish	3.7	1/
Non-chlorine retentive resin	3.7	1/
pH	3.8	2811
Shrinkage (warp and filling)	3.9	5556 3/
Seam efficiency	3.10	5110 4/
Sulfur Content	3.4.1.1	2020

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.
2/ One determination shall be made from each sample unit and the result shall be reported as "pass" or "fail".
3/ Cotton procedure shall be used.
4/ The needle shall measure 0.044 inch (± 0.001 inch) across the blade at the eye. The thread shall be either polyester/cotton in accordance with MIL-T-43548, ticket No. 50, 2 ply for the needle and ticket No. 70, 2 ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or class 2, ticket No. 40, 3 ply for the needle and ticket No. 70, 2 ply for the looper.
5/ The specimens must be dried after each of the 3 laundering cycles.
6/ Only the stain on the cotton and polyester fibers of the color transfer cloth shall be evaluated.

4.5.1 Test for polyester fiber and cotton fiber content. The general procedures of test method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}} = \text{percent polyester}$$

100 - percent polyester = percent cotton

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 0.1 percent.

5. PACAKAGING

5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of uniform clothing items used by Naval personnel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Class and color of cloth required (see 1.2 and 3.4.1).
(c) When first article sample is required (see 3.2).
(d) Width of cloth required (see 3.5.1).
(e) Length of roll required if other than specified (see 3.11).
(f) Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Fabric Defect Replica Kit", Sears Tower, Chicago, Illinois 60684.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29127A	Cloth, Twill, Polyester/Cotton (Crease-Resistant Finish)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	8. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	9. DATE OF SUBMISSION *(YYMMDD)*

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